

Artavazd Maranjyan

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Current Position

2026–present **Postdoctoral Researcher, working with Volkan Cevher and Martin Jaggi**, *EPFL*, Lausanne, Switzerland

Education

- 2023–2025 **Ph.D. in Computer Science**, *KAUST*, Thuwal, Saudi Arabia; **Advisor:** Peter Richtárik
■ **Thesis:** [First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data](#)
- 2021–2023 **M.Sc. in Applied Statistics and Data Science**, *Yerevan State University*, Yerevan, Armenia
■ **Thesis:** [On local training methods](#); **Co-supervisors:** Peter Richtárik, Mher Safaryan
- 2017–2021 **B.Sc. in Informatics and Applied Mathematics**, *Yerevan State University*, Yerevan, Armenia
■ **Thesis:** [On the Convergence of Series in Classical Systems](#); **Supervisor:** Martin Grigoryan

Selected Industry and Academic Experience

- 03/2025 **Research Visit to Yi-Shuai Niu**, *BIMSA*, Beijing, China
○ Gave talks at three universities (PKU, BUAA, BIMSA)
○ Collaborated with Professor Yi-Shuai Niu on a project on Server-Assisted Federated Learning
- 06/22–01/23 **Internship in the Group of Peter Richtárik**, *KAUST*, Thuwal, Saudi Arabia
○ Worked on the "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity" paper
- 07/21–06/22 **Co-Founder**, *OnePick*, Yerevan, Armenia
○ Built an MVP platform for generating or selecting photos matching text descriptions
○ Participated in the [InVent 2.0 \(FAST\)](#) venture building program, from idea generation to pitching
○ Secured initial funding and continued development beyond the program
- 09/19–01/21 **Software Engineer in Test**, *Picsart*, Yerevan, Armenia
○ Worked with automation team to design and develop automated solutions across several mobile/web applications
○ Worked with software developers, test engineers, product owners, business analysts to find and resolve issues
○ Worked closely with DevOps to suggest improvements in processes and in Jenkins Continuous Integration cycle

Awards

- 06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowships**, *EPFL*
Competitive fellowship supporting independent postdoctoral research, with additional travel funding
- 05/2025 **CEMSE Dean's List Award**, *KAUST*
Awarded for excellent academic and research performance (\$2,500 prize)
- 09/2023 **Dean's Award**, *KAUST*
Awarded to a few top students accepted to KAUST (\$6,000 annually for 3 years)
- 05/2021 **Outstanding Final Project Award**, *Yerevan State University*
Recognized for the Bachelor's thesis (awarded to 6 students among 250+ students)

Selected Publications

7. **Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method**
Abdurakhmon Sadiev, [Artavazd Maranjyan](#), Ivan Ilin, Peter Richtárik
arXiv:2605.18174, 2026
6. **Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity**
Ammar Mahran, [Artavazd Maranjyan](#), Peter Richtárik
arXiv:2605.13434, 2026
5. **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity**
[Artavazd Maranjyan](#), Peter Richtárik
ICLR 2026: The Fourteenth International Conference on Learning Representations

4. **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity**
Artavazd Maranjyan, Alexander Tyurin, Peter Richtárik
ICML 2025: Forty-Second International Conference on Machine Learning
3. **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning**
Artavazd Maranjyan, El Mehdi Saad, Peter Richtárik, Francesco Orabona
ICML 2025: Forty-Second International Conference on Machine Learning
2. **MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
Artavazd Maranjyan, Omar Shaikh Omar, Peter Richtárik
UAI 2025: The 41st Conference on Uncertainty in Artificial Intelligence
OPT 2024: Optimization for Machine Learning (NeurIPS workshop)
Oral presentation (top 5% of 107 submissions)
1. **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [\[video\]](#)
Artavazd Maranjyan, Mher Safaryan, Peter Richtárik
TMLR 2025: Transactions on Machine Learning Research

Reviewing

- 2024–present TMLR (Transactions on Machine Learning Research)
- 2026 NeurIPS (Conference on Neural Information Processing Systems)
- 2026 ICML (International Conference on Machine Learning)
- 2025 ICLR (International Conference on Learning Representations)
- 2024 SIMODS (SIAM Journal on Mathematics of Data Science)
- 2024 JMLR (Journal of Machine Learning Research)

Selected Talks

- 26/04/2026 **Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)**, Lagune Barra Hotel, Rio de Janeiro, Brazil
Delivered a talk titled **Controlling Delay in Asynchronous SGD: Optimality for Any Data Regime** [\[video\]](#) [\[slides\]](#)
- 29/07/2025 **35º Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil
Delivered a talk and a poster on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[slides\]](#) [\[poster\]](#)
- 16/04/2025 **FLOW Talk #126**, Federated Learning One World Seminar (FLOW), Online
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[video\]](#) [\[slides\]](#)
- 17/03/2025 **Academic Report of the School of Mathematical Sciences**, Beihang University (BUAA), Beijing, China
Invited by Jiaxin Xie to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[slides\]](#)
- 13/03/2025 **Optimization Seminar**, BIMSA, Beijing, China
Invited by Yi-Shuai Niu to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[slides\]](#)
- 12/03/2025 **Seminar**, Peking University, Beijing, China
Invited by Kun Yuan to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[slides\]](#)
- 21/11/2024 **MLR Weekly Seminar**, Machine Learning Research at Apple, Online
Invited by Samy Bengio to give a talk on **MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [\[slides\]](#)
- 13/03/2024 **The Machine Learning Summer School in Okinawa 2024**, OIST, Okinawa, Japan
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [\[poster\]](#)
- 06/10/2023 **Algorithms & Computationally Intensive Inference seminars**, University of Warwick, Coventry, England
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [\[slides\]](#)